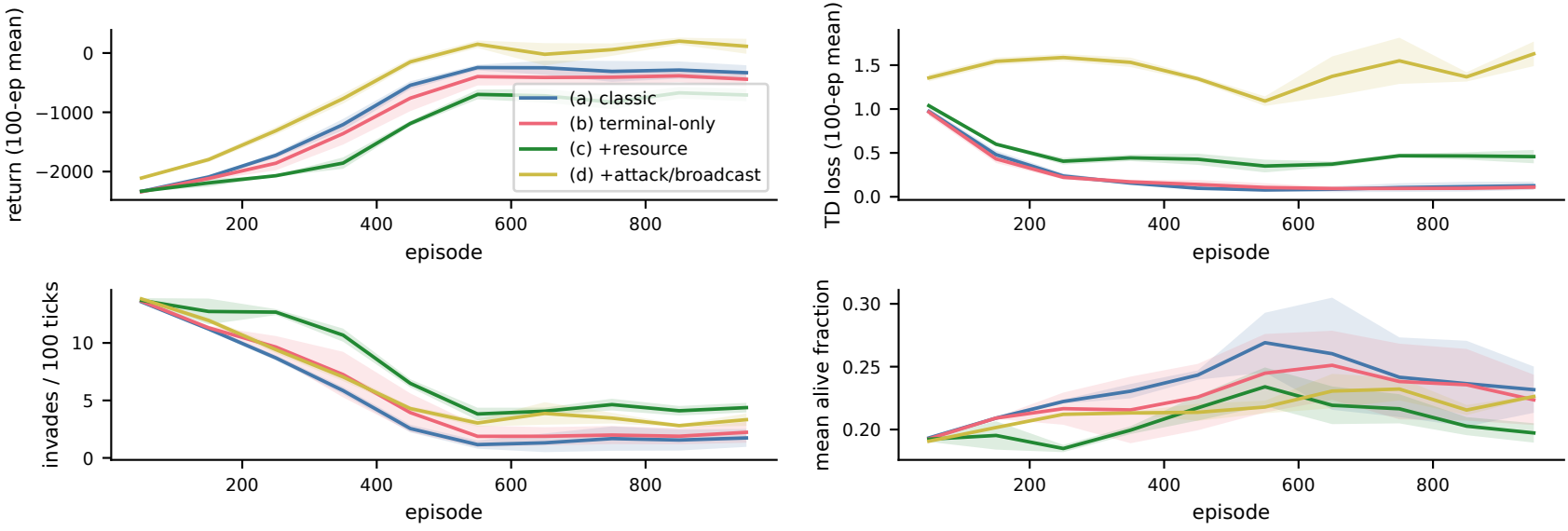
Training convergence by reward arm (v2.5, 3 seeds, mean±sd)
return (100-ep mean)
0
−1000
−2000
200
400
600
800
episode
(a) classic
(b) terminal-only
(c) +resource
(d) +attack/broadcast
TD loss (100-ep mean)
1.5
1.0
0.5
0.0
200
400
600
800
episode
invades / 100 ticks
10
5
0
200
400
600
800
episode
mean alive fraction
0.30
0.25
0.20
200
400
600
800
episode